

Business Requirements Document

Motorcycle Parts & Accessories Business Management System

Version: V1.0 **Phase 1:** Internal system **Phase 2:** Customer website

1. Executive Summary

A centralized system for a three-partner motorcycle-parts importing business. It replaces Excel/WhatsApp/order-history tracking and manages import cycles, purchases, shipping, inventory batches, B2B/B2C sales, customer balances, partner/investor finance, analytics, notifications and audit logs.

2. Business Model

- Three permanent core partners have equal authority. Contributions normally match but may vary by cycle.
- Products are selected first; required capital is then calculated.
- One physical shipment = one Import Cycle. A cycle may contain multiple Alibaba/supplier orders.
- Products are mainly sourced from China, sometimes directly from UAE.
- China→UAE is handled by the merchant. UAE→Egypt is handled by a shipping company with one combined payment covering its services/customs/handling.
- Current physical inventory is stored in a room at home.
- Sales are mainly B2B to motorcycle shops, with B2C sales through channels such as Facebook Marketplace.

3. Scope

Phase 1 — Internal: Dashboard, cycles, purchases, suppliers, shipments, products, inventory, sales, customers, payments, finance, partners/investors, analytics, notifications and audit logs.

Phase 2 — Customer portal: public catalog, verified shop accounts, B2B ordering/requesting, out-of-stock product requests and customer history.

4. Users & Authority

User	V1 access
Core Partner A/B/C	Equal full internal authority; every important action is logged.
Temporary Investor	Cycle-specific financial participant; no internal authority by default.
Shop Owner (Phase 2)	Verified account; sees own products/orders/requests/balances only.

5. Import Cycle

- Workflow: Planning → select products → estimate capital → agree contributions → purchase → logistics → receive in Egypt → verify inventory → sell → collect payments → calculate final profit → settle → close.
- Suggested statuses: Planning, Funding, Purchasing, In Transit, Arrived UAE, In Transit to Egypt, Arrived Egypt, Verification, Selling, Settlement, Closed.
- China→UAE and UAE→Egypt are separate logistics legs. UAE-direct purchases use only UAE→Egypt.
- Products become saleable inventory after arrival and verification in Egypt.

6. Products & Inventory

- Internal SKU generated by the system; Alibaba/supplier identifiers and links stored separately.
- Product data: name, category, description, photos, motorcycle compatibility, suppliers, B2B price, B2C price, minimum stock.
- Same product may have multiple suppliers.
- Barcode is out of scope for V1 but data model should allow it later.
- Current physical location: Home Storage Room.
- Track total, reserved and available stock plus full stock-movement history.
- CRITICAL: inventory is batch-based. The same product from three cycles with different costs remains three distinguishable batches.

Example: Honda CBR Brake Pad

Cycle	Qty	Landed cost/unit	Remaining
#021	100	90 EGP	20
#024	150	110 EGP	100
#027	200	125 EGP	200

Proposed V1 costing: FIFO (oldest batch first). Each sale must retain the batch/cost used for COGS and profit calculation. Final confirmation remains a TBD.

7. Purchasing & Logistics

- Purchases belong to a cycle and may be multiple supplier/Alibaba orders.
- Record supplier, order reference, products, quantities, prices, currency, order date, refund/adjustment data and links.
- Shipping leg records origin, destination, provider, reference/tracking, dates, status and total amount paid.
- No separate customs expense in the normal workflow; the UAE→Egypt shipping payment is recorded as one amount.
- Supplier refunds are linked to the original purchase.

8. Partner & Temporary Investor Finance

- Default cycle profit distribution is proportional to actual contribution.
- Example: A 80k, B 100k, C 120k → profit shares 26.67%, 33.33%, 40%.
- Support explicit custom profit distribution when the partners agree to an exception.
- Temporary investors participate only in specified cycles.
- Temporary investor's gross profit is calculated from the cycle agreement. An agreed percentage is deducted from that investor's profit, not capital. Example: 50k gross profit × 15% = 7.5k fee; investor keeps 42.5k profit.
- Capital return and profit are separate in settlement.

9. Sales & Customers

- Support B2B and B2C sales, customer history, orders, partial payments and outstanding balances.
- Support Sunday installment payments; exact overdue/installment rules remain TBD.
- Support stock reservation.
- Returns/refunds/adjustments must preserve history.
- Phase 2 shop owners must log in before creating requests/orders.

10. Finance & Ledger

- Track partner contributions, product purchases, shipping, other expenses, customer payments, refunds, reimbursements, withdrawals and settlements.
- Support multiple money accounts later.
- Every money movement has amount, currency, date, category, related cycle/entity where applicable and creator.
- Do not hard-delete financial records; use reversal/cancellation.

11. Analytics & Reporting

- Dashboard: revenue, profit, expenses, inventory value, receivables, active cycles and alerts.
- Compare by day/week/month/quarter/year/custom period.
- Compare cycles/orders: investment, landed cost, revenue, profit, ROI, units purchased/sold and remaining stock.
- Product: units, revenue, profit, margin, fast/slow movers.
- Customer: purchases, payments, outstanding balances and trends.
- Partner/investor: contribution and profit by cycle, respecting permissions.
- Sales channel: B2B vs B2C and future Amazon/Jumia/website channels.
- Cash flow must be shown separately from accounting profit.

12. Notifications

- Low stock, new product request, payment due/overdue, shipment changes, arrival/verification required, refunds, cycle settlement, investor settlement and other important events.
- V1: in-app notification center. Email/WhatsApp/push can be added later.

13. Audit Logs

- Record who performed important actions and when.
- For edits, retain old/new values where practical.
- Log financial changes, inventory movements, prices, agreements and status changes.
- All three core partners have equal authority, but actions remain attributable to the user.

14. Phase 2 Customer Website

- Public catalog browsing without login.
- Login/verified shop account required for requests and orders.
- Shop sees B2B pricing, own orders, requests, payment history and balance.
- Out-of-stock/new-product request captures product name, bike compatibility, quantity, notes and optional photo/Alibaba link.
- Customer requests become demand data that can be attached to future cycles.
- Never expose internal costs, supplier information, partner contributions or profit agreements.

15. Suggested Screens

- Dashboard; Cycles; Purchases; Shipments; Products; Inventory; Sales; Customers; Payments/Receivables; Expenses/Ledger; Partners/Investors; Analytics; Notifications; Audit Logs; Settings.

16. Suggested Technical Architecture

- Frontend: Next.js + TypeScript (App Router).
- Backend: NestJS + TypeScript.

- Database: PostgreSQL + Prisma.
- Product photos: object storage.
- API-first architecture so Phase 2 customer website uses the same backend.
- Use database transactions and precise decimal types for financial/inventory operations.

17. Suggested Core Entities

Entity	Purpose
User / Partner / Investor	Identity and participation
ImportCycle / CycleParticipant	Shipment lifecycle and financial agreement
Supplier / PurchaseOrder	Sourcing
ShippingLeg	Logistics and payment
Product / ProductSupplier / Compatibility	Catalog
InventoryBatch / InventoryMovement	Cycle-specific stock and cost
Customer / SaleOrder / SaleItem / Payment	Sales and receivables
Expense / FinancialTransaction	Money ledger
InvestmentAgreement / Settlement	Temporary-investor terms and distributions
ProductRequest	Future customer demand
Notification / AuditLog	Alerts and traceability

18. Key Business Rules

- One shipment is one cycle; a cycle can contain multiple purchases.
- Profit distribution defaults to proportional to actual contribution.
- Custom profit distribution can be agreed per cycle.
- Temporary-investor fee is a percentage of that investor's allocated profit.
- Same product can have multiple cycle-specific batches and costs.
- Historical prices/transactions must not be overwritten.
- Supplier refunds are linked to original purchases.
- Capital and profit are separate settlement components.
- Financial and inventory history should not be hard-deleted.

19. Open Decisions (TBD)

- Exact Sunday installment and overdue rules.
- Customer return/refund policy.
- How to settle a cycle when inventory remains.
- Whether leftover stock can later be transferred from a cycle to general inventory.
- How the core partners split a temporary investor's fee (equal is currently assumed, not locked).
- Final inventory costing method (FIFO proposed).
- Exact payment-account/reconciliation workflow.
- External notification channels.

20. MVP Acceptance Criteria

- Create a cycle with purchases, expected costs and participant contributions.
- Support China→UAE→Egypt and UAE→Egypt routes.
- Receive and verify inventory into cycle-specific batches.
- View the same product across cycles with different costs.
- Record sales and reduce the correct inventory batch.
- Calculate revenue, COGS, profit and proportional participant shares.
- Apply temporary-investor fee to allocated profit.
- Track customer payments and outstanding balances.
- Record expenses/refunds.
- Provide analytics, comparisons, notifications and audit history.

21. Recommended Next Step

Before coding, convert this BRD into a technical specification: finalized business rules, ERD/database schema, API contracts, page-level requirements, permissions, financial formulas and test scenarios. This is the document that should then be handed to OpenCode/Big Pickle as the implementation source of truth.

Status: V1.0 working BRD based on the requirements discussed so far. TBD items should be confirmed before production implementation.